

SendCUIEmail Documentation

SendCUIEmail

Encrypt files for secure email transmission when certificate exchange isn't possible. Designed for CUI (Controlled Unclassified Information) handling per NIST SP 800-171.

Security Hierarchy

Rank	Method	When to Use
1	S/MIME (PIV/CAC to PIV/CAC)	Both parties have smart cards. Use Outlook's native Sign+Encrypt. Gold standard.
2	SendCUIEmail + S/MIME password	Recipient has PIV/CAC. S/MIME encrypt the password email.
3	SendCUIEmail + out-of-band password	No PIV/CAC. Share password via phone/SMS/in-person.
4	SendCUIEmail + unencrypted email password	NOT COMPLIANT - convenience only with warnings.

When to Use This Tool

Use native S/MIME (don't need SendCUIEmail) when: - Both parties have PIV/CAC with valid certificates - Recipient's certificate is accessible - Regular communication with same recipients

Use SendCUIEmail when: - Recipient does not have PIV/CAC card - Recipient's certificate is not accessible (different organization) - One-time or infrequent communication with external parties - Recipient's email client does not support S/MIME - Files need to be forwarded/stored on non-S/MIME systems

See [Decisions/DM-2026-003_password_transmission.pdf](#) for full guidance.

Features

- **FIPS 140-2 Compliant:** AES-256-CBC encryption with PBKDF2-HMAC-SHA256 key derivation
- **CUI Marking Support:** Built-in CUI category selection per 32 CFR Part 2002
- **No Dependencies:** Uses only built-in Windows PowerShell cryptography
- **Email Ready:** Generates Outlook .msg draft with proper CUI banners and attachments
- **Recipient Friendly:** Recipients decrypt with a single PowerShell command (no executables)

Quick Start

Encrypt Files

1. Drag files or folders onto **Encrypt.bat**
2. Select CUI category
3. Enter password (twice to confirm)
4. Send the generated files to recipient

Output: - *.Locked - Encrypted files - README.md - Decryption instructions for recipient - CUI_Email_*.msg - Outlook draft with encrypted files (if Outlook installed) - Password_Email_*.msg - Outlook draft with password (send separately or use alternate channel)

Decrypt Files

Recipients paste this PowerShell one-liner (from the README.md you send them):

```
$f=Read-Host "File";$p=Read-Host "Password" -AsSecureString;$d=[IO.File]::ReadAllBytes($f);$k=
```

Or use **Decrypt.bat** if they have the tool.

Files

File	Purpose
Encrypt.bat	Drag-and-drop encryption launcher
Encrypt.ps1	Main encryption script
Decrypt.bat	Drag-and-drop decryption launcher
Decrypt.ps1	Decryption script
Test.bat	Run round-trip verification tests
Test.ps1	Test script

What Gets Sent to Recipient

Email 1 (Encrypted Files): - OriginalName.ext.Locked - Encrypted file(s) - README.md - Decryption instructions

Email 2 (Password) - Send separately, or better yet, use alternate channel: - Password only (no attachments)

NOT sent: - No .exe files - No .ps1 scripts - No .zip bundles - No .bat files

Password Transmission Options (per NIST SP 800-63B):

Method	Compliant?	Notes
Phone/SMS/in-person	Yes	True out-of-band channel
S/MIME encrypted email	Yes	Encrypt password email with recipient's PIV/CAC certificate
Unencrypted email	No	Even separate email is not compliant

Hybrid Approach: If recipient has PIV/CAC, S/MIME encrypt the password email (click “Encrypt” in Outlook). The password is then protected by asymmetric encryption - only the recipient can decrypt it.

See [Decisions/DM-2026-003_password_transmission.pdf](#) for full guidance.

File Size Limit

Maximum supported attachment size: 10 MB

Files larger than 10 MB should not be sent via encrypted email due to: - Email gateway size limits (typically 10-25 MB) - Base64 encoding overhead (~33% size increase) - Delivery reliability concerns - Mailbox quota limitations

For larger files, use: - Secure file transfer services (SFTP, approved cloud storage) - FIPS 140-2 validated encrypted USB drives - DoD SAFE, MilSuite, or similar approved platforms

See [Decisions/DM-2026-002_file_size_limit.pdf](#) for full rationale.

Security

Encryption Details

Parameter	Value
Algorithm	AES-256-CBC
Key Derivation	PBKDF2-HMAC-SHA256
Iterations	100,000
Salt	128-bit random
IV	128-bit random

Compliance

- **FIPS 140-2:** AES-256-CBC is a FIPS-approved algorithm
- **NIST SP 800-132:** PBKDF2 key derivation meets NIST recommendations
- **NIST SP 800-171:** CUI handling requirements
- **32 CFR Part 2002:** CUI marking requirements

CUI Categories Supported

1. CUI - Basic Controlled Unclassified Information
2. CUI//SP-CTI - Controlled Technical Information
3. CUI//SP-EXPT - Export Controlled
4. CUI//SP-PRVCY - Privacy
5. CUI//SP-PROPIN - Proprietary Business Information

Testing

Automated Tests

Run the test suite to verify encryption/decryption:

```
# Windows
.\Test.bat
```

```
# macOS/Linux
./test.sh
```

Test Data Files

Pre-built test files are in `testdata/`:

File	Size	Purpose
<code>small_text.txt</code>	138 B	Basic text file
<code>empty.txt</code>	0 B	Edge case: empty file
<code>binary_sample.bin</code>	1 KB	Binary with null bytes
<code>medium_file.bin</code>	1 MB	Medium binary file
<code>large_file_10mb.bin</code>	10 MB	Maximum size test
<code>README.pdf</code>	~175 KB	Generated from README.md
<code>README.docx</code>	~15 KB	Generated from README.md
<code>README.png</code>	~200 KB	First page of README as image
<code>sample.xlsx</code>	~5 KB	Excel spreadsheet format

Generate test files (if missing):

```
./testdata/generate_testdata.sh
```

Sample LaTeX Document

A LaTeX source document is in `examples/test_document.tex`. Compile to PDF:

```
cd examples
pdflatex test_document.tex
```

Requirements

Windows (Primary Target): - Windows 10/11 - PowerShell 5.1 or later (included with Windows)
- Microsoft Outlook (optional, for .msg generation)

macOS/Linux (Development/Testing): - PowerShell Core 7+: `brew install powershell` (macOS) or see [Microsoft docs](#) - Outlook .msg generation not available (gracefully skipped) - Core encryption/decryption fully functional

Usage Examples

Single File

```
Encrypt.bat document.pdf
```

Multiple Files

```
Encrypt.bat file1.pdf file2.docx file3.xlsx
```

Entire Folder

```
Encrypt.bat "C:\Sensitive\Documents"
```

Command Line (PowerShell)

```
powershell -ExecutionPolicy Bypass -File Encrypt.ps1 "document.pdf"
```

Troubleshooting

“Script cannot be loaded” error

Run via the .bat file, or set execution policy:

```
Set-ExecutionPolicy -Scope CurrentUser -ExecutionPolicy RemoteSigned
```

Outlook .msg not generated

This is optional - Outlook must be installed. You can manually attach files to email.

Decryption fails

- Verify password is correct
- Ensure file wasn't corrupted during transfer
- Check file has .Locked extension

License

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Related Projects

- [PDFSigner](#) - Digital signature tool for PDFs